Glaucoma Detection Using YOLO and ResNet50 Deep Learning Model

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1 Introduction

Glaucoma is a serious eye disease that damages the optic nerve and can lead to permanent vision loss if not detected early. It often develops without noticeable symptoms, making early diagnosis difficult.[1] Traditional detection methods rely on clinical examinations and expert analysis, which can be time-consuming. Recently, artificial intelligence especially deep learning has shown strong potential for automated detection using retinal fundus images. This project focuses on improving detection accuracy by analyzing only the optic disc region, the most relevant area for diagnosis.

2 Methodology

The conventional approach for glaucoma detection involves training models using full retinal fundus images. However, this approach introduces unnecessary noise and can negatively affect the model's ability to focus on important features.[2] To address this limitation, a two-stage methodology is proposed, **Stage 1: Optic Disc Detection** and **Stage 2: Glaucoma Classification**. This approach ensures that only the most relevant region, the optic disk, is used for classification, thereby improving model efficiency and performance.

2.1 Optic Disc Detection using YOLO

A YOLO object detection model is used to locate the optic disk region in retinal images.The full dataset consists of **748 retinal fundus images**. However, due to time and resource constraints, only **120 images were annotated** for optic disc detection. Despite the reduced labeling size, this subset was sufficient to train a reliable detection model. The optic disk is a visually distinct and consistent anatomical structure across retinal images, which allows the model to generalize effectively even with a smaller labeled dataset. The labels are created manually using LabelImg, which generates XML annotation files containing bounding box coordinates. These annotations are used to train the YOLO model for optic disc localization.

2.1.1 Image Preprocessing

To improve detection performance, image enhancement techniques are applied prior to training, including **CLAHE** to enhance contrast and highlight important structures, **grayscale conversion** to simplify image features and reduce computational complexity, and **Gaussian blur** to reduce noise and smooth the image. These preprocessing steps improve the visibility of the optic disc and help the model detect it more accurately.[3]

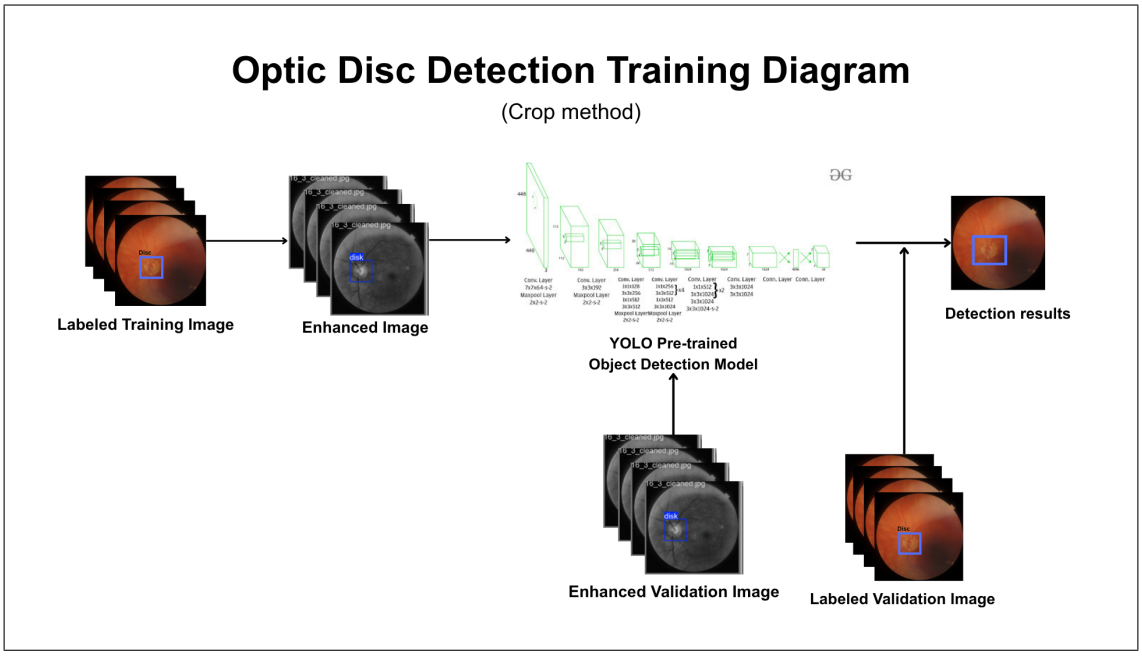

Figure 1: Optic Disc Detection Training Diagram

2.2 Optic Disc Cropping

After training, the YOLO model is used to detect and crop the optic disc from each retinal image. This step removes irrelevant regions other than optic disc to reduce unnecessary information, ensuring that the classification model focuses only on the most important area for glaucoma detection.

2.3 Glaucoma Classification using ResNet50

The cropped optic disc images are used as input to a ResNet50 model that will analyse the cropped image and performs binary classification to **Class 0: Normal** and **Class 1: Glaucoma**. The model ResNet50 is selected due to its deep architecture and ability to extract complex features from medical images.[4] By using cropped images instead of full images, the model is able to Reduce noise from irrelevant regions, focus on clinically significant features and improve learning efficiency

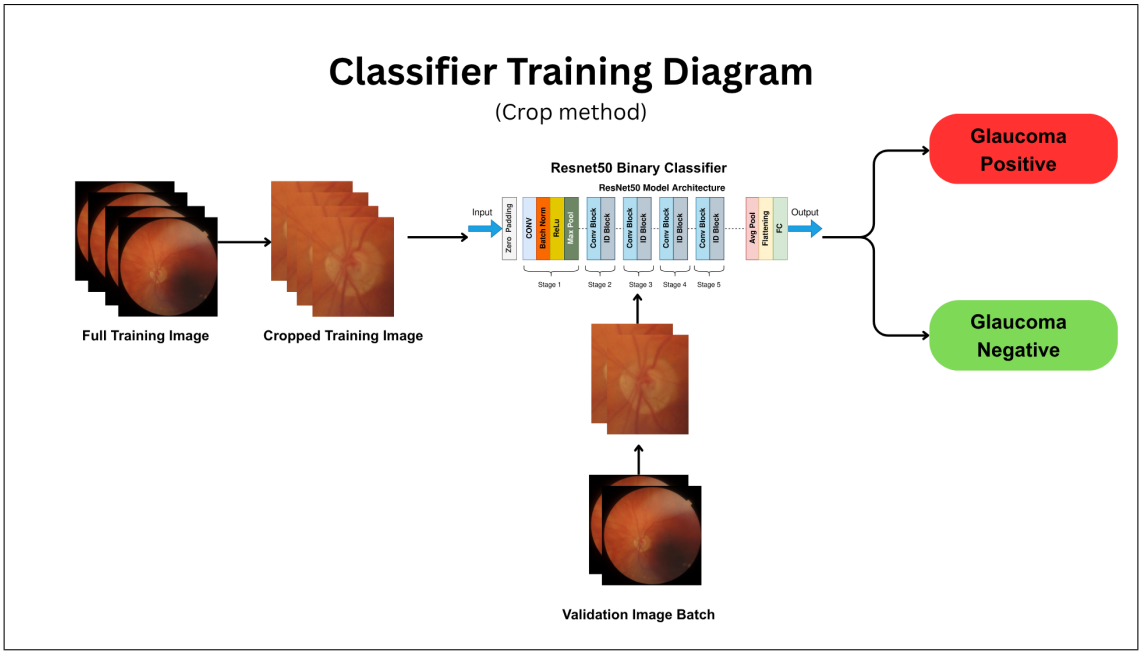

Figure 2: Classifier Training Diagram

3 Data Analysis

3.1 Part 1: Optic Disc Detection

The YOLO model was trained to detect the optic disc with high precision, showing steady decreases in training loss and similar trends in validation loss, indicating good learning without significant overfitting. At the same time, mAP50 improves over epochs, reflecting increasing detection accuracy. Here, loss represents prediction error (lower is better), with train loss measured on the training dataset and validation loss on the validation dataset, while mAP50 evaluates detection accuracy at an IoU threshold of 0.5.

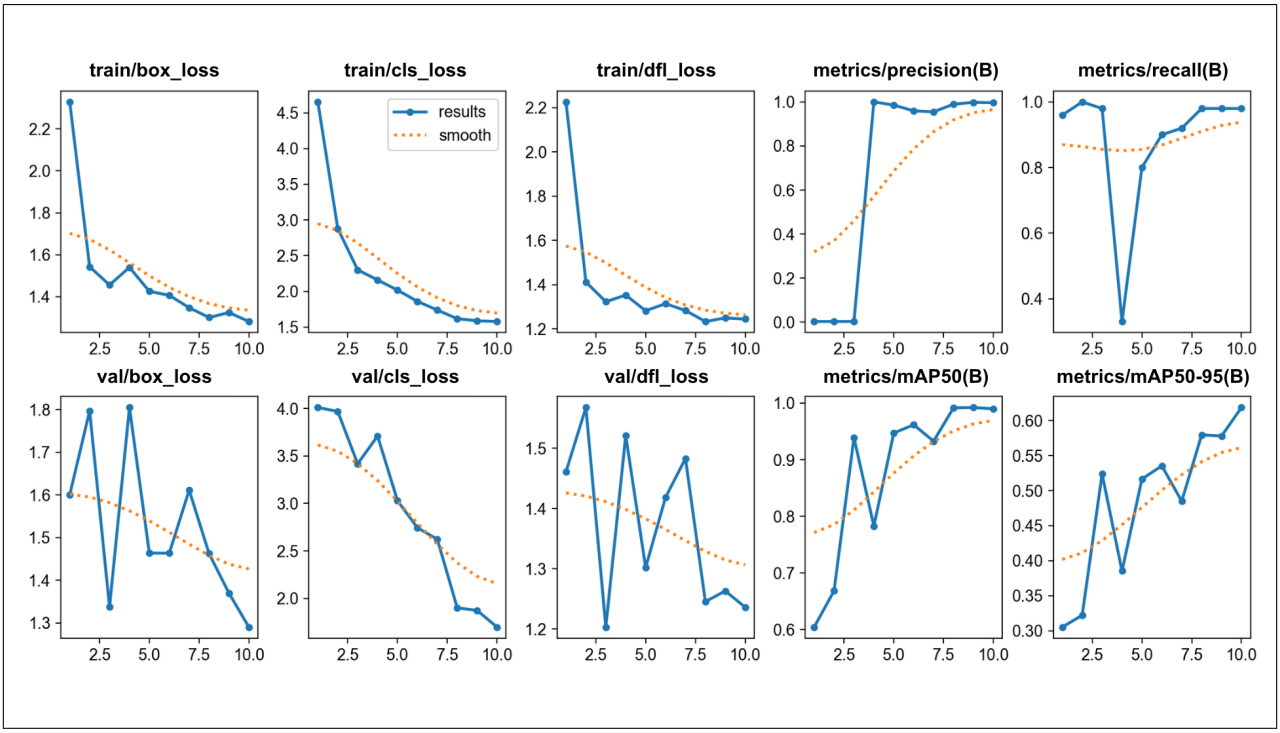

Figure 3: YOLO Training Metrics

The model achieves approximately **98% accuracy** in detecting the optic disc with stable confidence scores.

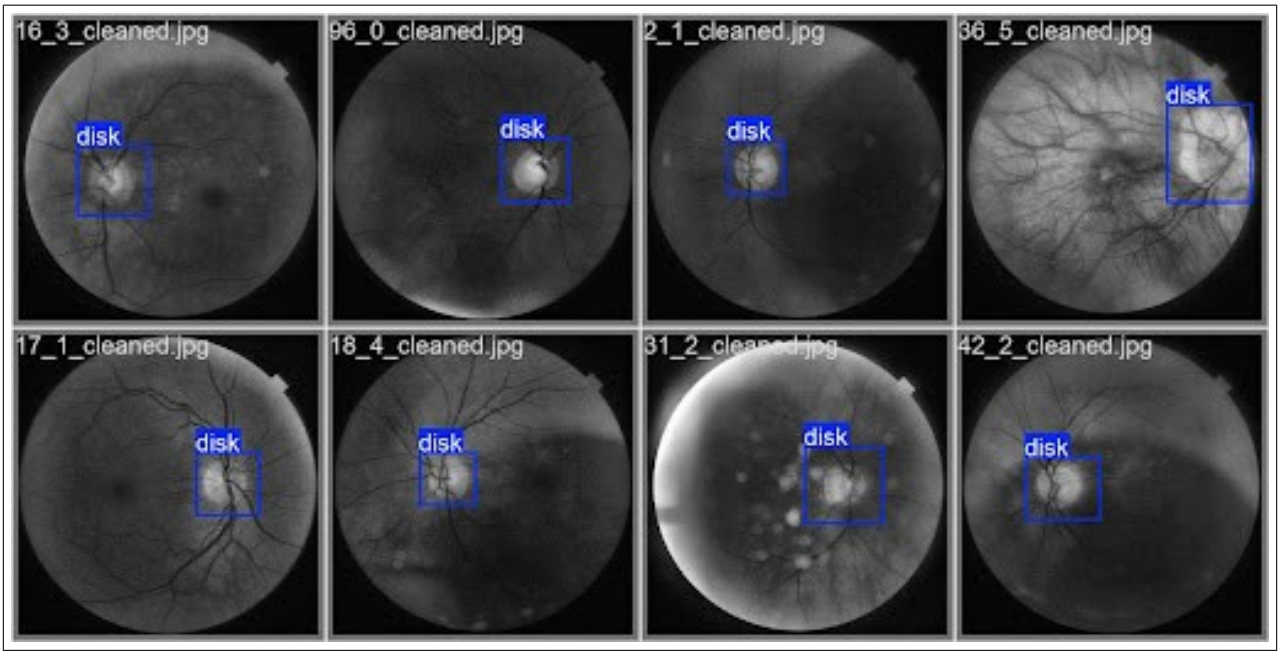

Figure 4: Optic Disc Detection Results

3.2 Part 2: Glaucoma Detection

The ResNet50 classifier is trained using cropped optic disc images.

Final evaluation results:

True Positive (TP)	True Negative (TN)	False Positive (FP)	False Negative (FN)
54	14	12	1

Performance Metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = 83.95\%$$
$$\text{Sensitivity (Recall)} = \frac{TP}{TP + FN} = 98.18\%$$
$$\text{Specificity} = \frac{TN}{TN + FP} = 53.85\%$$

Interpretation:

High sensitivity indicates that the model is very effective at detecting glaucoma cases, while the lower specificity suggests that some normal cases are incorrectly classified as glaucoma. However, for medical screening applications, higher sensitivity is often prioritized over specificity to reduce the risk of missing true disease cases.[5]

4 System Implementation

A web-based system is developed using Flask to demonstrate the prediction pipeline. The process begins when the user uploads a retinal fundus image. The YOLO model then detects the optic disc within the image, after which the optic disc is cropped. The cropped image is subsequently passed to the ResNet50 model for analysis, and finally, the system displays the prediction indicating whether the case is glaucoma or normal.

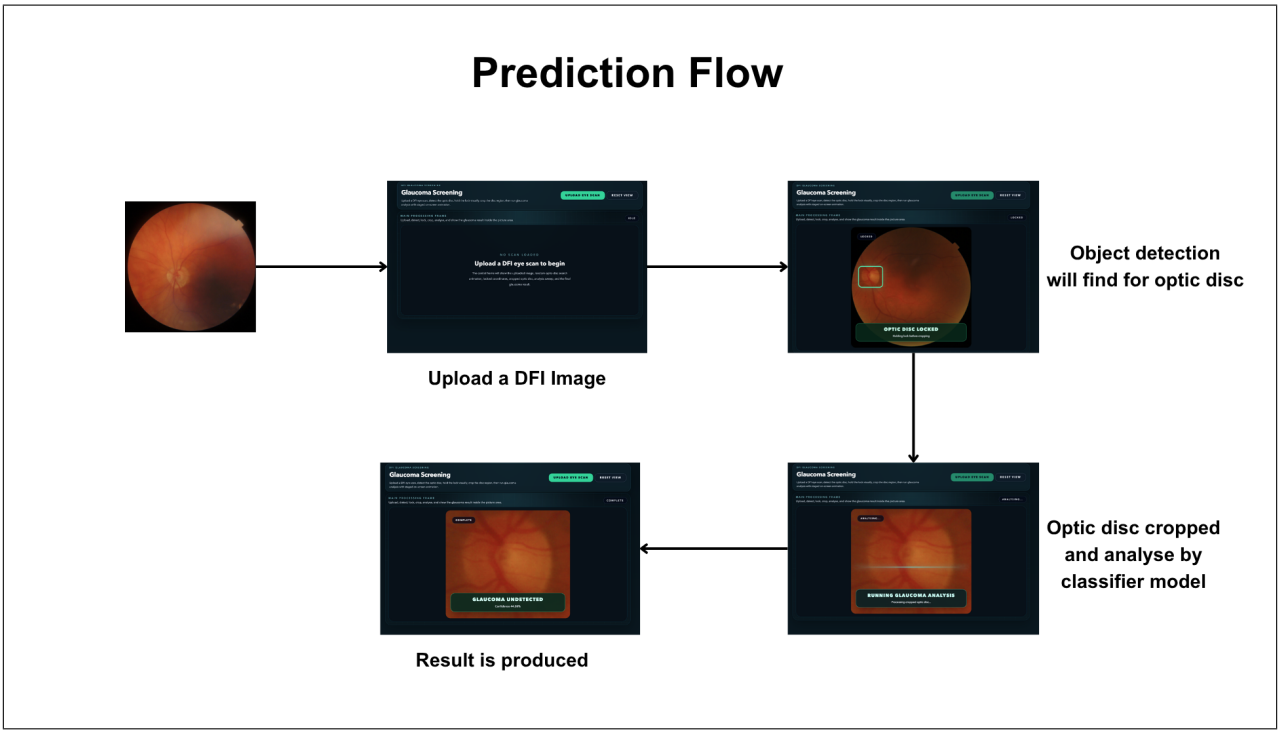

Figure 5: Prediction Flow in Web Application

5 Limitations and Challenges

The dataset exhibits an imbalance, with more glaucoma images than normal images, which can introduce bias into the model. To address this, class weighting is applied to reduce the impact of this imbalance, while future work may involve using data augmentation techniques such as rotation and flipping to increase the number of normal samples. Additionally, the model has a relatively high computational cost, as training takes approximately 30 minutes per run and fine-tuning requires multiple iterations; this can be mitigated by using GPU acceleration to achieve faster training.

6 Conclusion

This project demonstrates an effective approach for glaucoma detection by combining object detection and classification models. By focusing on the optic disc region instead of the full image, the model reduces noise and improves its ability to learn relevant features. The YOLO model successfully detects the optic disc with high accuracy, while the ResNet50 classifier achieves strong sensitivity in detecting glaucoma cases. Although the model shows promising results, improvements can be made in handling dataset imbalance and increasing specificity. Future work includes applying advanced data augmentation techniques and optimizing model architecture.

References

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